

A Class–Lab–Field Pedagogical Framework for Structural Health Monitoring Using Ultralow-Cost Wireless IoT Prototypes

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Abstract—Contribution: A Class–Lab–Field (CLF) framework for structural health monitoring (SHM) education centers on ArduinoNode, an open, ultralow-cost (~U.S. \$50/node) wireless Internet of Things (IoT) platform students build, program, and deploy, narrowing the analysis–field gap from sensing to interpretation.

Background: SHM courses foreground structural models and algorithms over instrument working principles and operational procedures; many Arduino labs lack coordinated outdoor deployment. Inspectable, affordable kits for teaching instrument internals remain scarce.

Intended outcomes: Operational literacy along the SHM pipeline: accelerometer fundamentals, instrument limits, and links among acquisition, synchronization, logging, and defensible interpretation of noisy field data.

Application design: CLF sequences classroom node preparation, lab work on a structural model, and field multisystem monitoring for compressed programs. Hands-on sensing comes first; SHM theory depth scales with contact hours.

Findings: CLF: APESS2025 ($N = 56$): 93.3% satisfaction, 100% recommend; ~60% report sensor/SHM/IoT gains, 40% WSN gains, and 60% rate hands-on emphasis much or significantly above typical theory-heavy courses. ArduinoNode: Footbridge data (ArduinoNode, smartphones, commercial SHM) compared with a finite-element (FE) model yield 5.1% and 7.2% errors on the first two modes under dual-link, sub-10-ms-class synchronization.

Index Terms—Class–lab–field (CLF) learning, engineering education, experiential learning, Internet of Things (IoT) platforms, structural health monitoring (SHM), wireless sensor networks.

I. INTRODUCTION

STRUCTURAL health monitoring (SHM) education sits at the intersection of civil performance assessment and instrumentation competence: graduates are expected to reason from noisy field data, yet many programs still under-supply how those data are produced. Professionally, SHM is practiced as a monitoring pipeline whose stages include acquisition,

preprocessing, storage, transmission, analysis, and decision-making, which must connect end-to-end [1], [2]. For pedagogy, that pipeline can be grouped roughly into data acquisition and data usage. Sound practice requires linking both; in practice, however, courses often foreground data usage (e.g., structural modeling, system identification, and interpretation, among others) while under-teaching data acquisition, so the upstream chain is frequently treated as an opaque model. Students may then see little of how sensing, timing, logging, and transmission shape the records on which later analysis depends, and may lack accelerometer-level judgment of data quality and instrument limits.

Beyond that curriculum-level pattern, curriculum critiques and engineering-education studies have long documented imbalances between theory and practice in civil programs, together with calls for stronger practical instruction [3], [4]. In parallel, reform efforts have emphasized project-based learning and capstone-style experiences that tie classroom learning to realistic tasks [5], [6]. The wider engineering-education literature supplies complementary levers: reviews of active learning [7], project-based treatments of infrastructure life-cycle roles [8], remote laboratory delivery [9], and constructionist practice in prototyping environments [10] each improve motivation and skills, but typically address engagement in bounded instructional formats rather than end-to-end monitoring chains. Bibliometric work likewise charts Arduino's diffusion across STEM education [11]. Taken together, these lines of work leave an opening for SHM: cohort-scale, field-coupled pipelines that rehearse deployable sensing and coordination across a class remain uncommon; modular SHM teaching efforts exist [12], yet scaling outdoor deployment and synchronized monitoring for an entire cohort is still not typical. Section II contrasts the present design with common project-based, remote-laboratory, and maker formats.

Instructional genres alone do not specify hardware; cohort-scale field work still requires nodes students can build, inspect, and synchronize. Civil engineering programs increasingly adopt diverse teaching and learning resources [13], but departments still need affordable kits students can operate, inspect, and tune; commercial SHM systems are often costly or closed, and access to research-grade hardware is uneven across institutions [14]. Open Arduino projects aimed specifically at SHM (as opposed to generic microcontroller exercises) remain sparse, and reported implementations often stop at introductory programming, single-node or single-

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channel demos, or bench-scale sensing rather than coordinated outdoor structural monitoring [15], [16], [17], [18], [19], [20]. Microcontroller memory, bandwidth, and power budgets limit sustained sampling, buffering, and logging, so industrial-grade SHM expectations on such hardware would face tight resource constraints; for introductory instruction, however, those limits can be acceptable when the goal is inspectable acquisition paths and classroom-scale coordination rather than asset-critical performance. Where Arduino appears as an accelerometer platform, explicit internode time synchronization in a teaching-oriented wireless sensor network remains uncommon, whereas service monitoring presupposes precise alignment and dependable logging [21], [22], [23]. The ArduinoNode stack developed in the present project builds on the Arduino UNO R4 Wi-Fi with native onboard IEEE 802.11 (no shield or stackable expansion boards required), pairs Wi-Fi with an nRF24L01 link layer, and implements synchronization that achieves sub-10-ms-class alignment in coordinated field capture (Section III). Educational SHM and Internet of Things (IoT) teaching accounts rarely report all of the following together: onboard 802.11 without add-on shields, student-visible firmware, and distributed timing at the precision needed for structural wireless labs. That gap is not a claim to have surveyed every paper; it is the reason the narrative foregrounds node architecture and synchronization mechanics rather than eliding acquisition.

Pedagogy and platform are paired deliberately. The instructional response is twofold: to stage learning along the monitoring chain introduced above, and to route students through inspectable hardware–software paths with repeatable operating procedures, supported by technology they can assemble, program, and deploy. Class–Lab–Field (CLF) sequences intensive classroom work on sensor assembly, embedded software, and bench operation; laboratory work on a structural model with reference acquisition for calibration and workflow rehearsal; and field deployment on a bridge so that sensing, logging, and comparison skills advance from bench to site under gradually reduced control. ArduinoNode is the open, inspectable platform carried from Class through Lab and Field: students assemble, flash, and bench-test the node in Class, then operate it in coordinated Lab experiments and Field deployment [16], [17], [24], [25]. A deliberate organizational pattern assigns each student team one node to assemble, program, and debug; Lab and Field then compose those nodes into a single class-wide wireless sensor network. Network size scales with the number of teams, while assembly and troubleshooting stay tractable because each group maintains only its own node rather than replicating a full multinode kit.

The CLF–ArduinoNode pairing above yields three main contributions: 1) a CLF instructional model for SHM sensor-network education (conceptual foundation, rationale for how stages are chosen and sequenced, and stage-to-activity mapping; Section II); 2) ArduinoNode as an embedded learning platform (architecture and mechanisms aligned to those stages; Section III); and 3) exploratory evaluation that combines questionnaires with technical feasibility checks, while noting limits of self-report data [26] (Section IV). Three guiding ideas (not formal hypotheses) organize the discussion: P1) CLF gives hands-on practice along the sensing and logging chain in

settings that open up step by step before advanced structural interpretation; P2) students should be able to adjust sampling, filtering, synchronization, and logging directly; and P3) comparing field-measured vibration spectra [e.g., Welch power spectral densities (PSDs) of on-bridge accelerations] to simple finite-element (FE) modal bands as interpretive targets builds interpretation skills that matter most when students move from the bench to the structure. Stronger evidence would need pre–post tests, scored student work, or hands-on performance tasks.

The remainder of this article is organized as follows. Section II presents the CLF framework: conceptual basis, summer-school staging, mapping of activities to practice and reflection, cohort context, and how implementation was carried out. Section III describes ArduinoNode (hardware, embedded software, time synchronization, and operating states) as used in Class, Lab, and Field. Section IV reports technical checks on the footbridge deployment and cohort-wide end-of-course questionnaire summaries ($N = 56$), presenting descriptive satisfaction and perceived-improvement patterns together with task-feasibility evidence. Section V closes with implications and directions for future work. Appendix URLs provide code, documentation, and the survey instrument.

II. CLF LEARNING FRAMEWORK AND EDUCATIONAL IMPLEMENTATION

A. Instructional Problem, Conceptual Foundations, and CLF Staging

Section I argued that SHM education should track the monitoring pipeline and that algorithm-centric instruction often under-supplies acquisition, field interpretation, and understanding of instrument internals (accelerometer paths, firmware, sampling, synchronization, logging) and operating procedures from assembly through deployment. Operators must run instruments correctly; understanding internals supports reasoned judgments about data quality and hardware envelopes that downstream analysis implicitly assumes. CLF and ArduinoNode address these gaps through Class exposition (assembly, flashing, communication protocols), structured Lab work, and authentic Field deployment. CLF supplies the instructional architecture; ArduinoNode (Section III) is the shared platform carried across Class, Lab, and Field: students assemble, flash, and bench-test it in Class, run coordinated experiments on it in Lab, and deploy it in Field, so the same artifact supports the full staged progression rather than entering only after the classroom phase.

CLF is organized around four ideas from the learning-science literature. Rather than stating them only in lecture, each idea is mapped to specific Class, Lab, and Field tasks so students repeatedly practice the same commitments in progressively harder settings. *Experiential learning*: Kolb's cycle [27], [28] informs the order of those tasks. In Class, students connect sensor and acquisition concepts to tangible work: assembling hardware, reading firmware structure, flashing code, and following operating procedures. In Lab, they run experiments on a structural model, reconstruct code, calibrate against reference sensing, and reflect on what the time series actually shows. In Field, emphasis shifts to deployment

logistics, fault handling, and comparing measured spectra to expectations. The cycle is used as a design rationale for sequencing, not as a psychometric instrument. *Authentic tasks*: Authentic learning stresses real-world context, coaching, and explaining one's reasoning [29]. The footbridge deployment is the anchor for that authenticity. Ambient noise, harmonics, and multipath effects are presented as signals students must interpret, not as defects to erase, consistent with professional monitoring practice. *Scaffolding and progressive disclosure*: Acquisition code is initially only partly visible; instructors guide reconstruction so that learners must justify sampling and filtering choices before seeing complete implementations. That pattern implements progressive formalization and counters “works without understanding” [30]. Prior work also notes that engineering students often favor laboratory work over lecture-only courses [31]; CLF routes that preference into an instrumentation-centered thread that stays tied to SHM deployment. *Epistemic practices*: Class links sensor principles to operating procedures. Lab adds a mechanism students can adjust and calibrate against References. Field requires interpreting data when excitation, hardware, or radio links are imperfect. Taken together, the three stages shrink the distance between textbook pictures of sensing and what real instruments output, including acquisition artifacts and the limits of the accelerometer signal chain.

Under compressed schedules typical of summer schools, Class time does not devote standalone blocks to SHM structural theory or system identification lectures. Instructors instead lead students through sensor hardware assembly, embedded programming and flashing, operating procedures, and simple bench checks so that the node and its firmware are inspectable before paired Lab work. Lab then moves to a structural model: teams run the ArduinoNode wireless sensor network while synchronized high-precision acquisition captures the same events for parameter calibration and for rehearsing acquisition workflows. Field shifts to a real footbridge: students practice deployment, including sensor placement, radio or link loss, data gaps, and coordination with other systems, alongside comparative analysis. The same progression can absorb additional contact hours (for example, SHM theory, richer analytics, or longer laboratory campaigns in Class and Lab when institutional schedules allow).

B. Course Design, Differentiation From Other Formats, and Conceptual Overview

The course does not reduce to a loose checklist of tasks. Practice, reflection, and analysis recur throughout Lab and Field. *Practice*: In Lab, students change acquisition settings on a structural model and run end-to-end workflows next to synchronized reference acquisition; in Field, they deploy on a bridge and interpret noisy traces with instructor coaching [27], [28], [29]. *Reflection*: Structured prompts repeatedly ask students to compare predictions with measurements in concrete situations, such as debugging traces, reviewing calibration residuals, reconciling power spectra, explaining spurious peaks, or addressing link and data dropouts. Reflection is thus scheduled rather than left implicit. *Analysis*: Multisensor comparison, progressive disclosure, and presentations ask learners

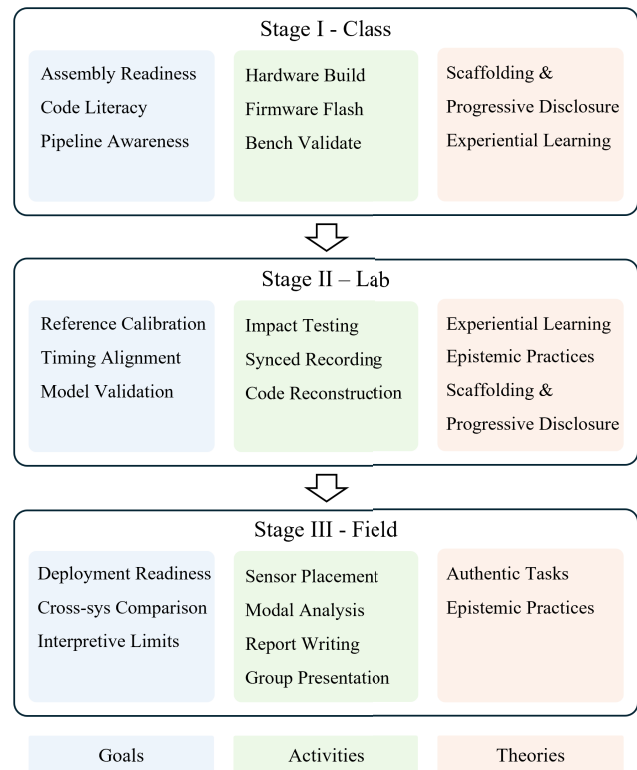


Fig. 1. Conceptual map of the CLF progression. Each stage lists goals, activities, and pedagogical theories (Section II-A). Photographic context is provided in the subsequent implementation figure.

to weigh data-quality tradeoffs and identification artifacts, not only to follow procedures. Together, these choices state the instructional logic plainly: each major activity ties to a named teaching idea so adopters can review and adapt the design.

The progression differs from project formats that omit a dedicated field phase [8] and from remote-laboratory setups that emphasize network access over co-located structural work [9], [32]. Maker spaces emphasize construction [10], [33]; CLF adds explicit alignment to the monitoring pipeline from acquisition through interpretation [7]. It is not a full replacement for those formats but combines hands-on construction with staged field deployment where SHM offerings often stop at the bench.

Fig. 1 summarizes CLF goals, activities, and theory emphasis across Class, Lab, and Field; Fig. 2 shows the concrete APES2025 implementation of that map in photographic form (Section II-C).

C. APES2025 Cohort and CLF Stages

The following describes the APES2025 cohort and how the staged offering was run. Fig. 2 provides photographic context for the same CLF progression summarized abstractly in Fig. 1: (a) and (b) instantiates Stage I (Class), (c) and (d) Stage II (Lab), and (e) and (f) Stage III (Field) from the goals-activities-theories map.

1) *Class Stage*: Class [Fig. 2(a) and (b)] concentrates on the sensing node itself: instructors supervise hardware assembly, embedded programming and flashing, bench-level checks, operating procedures, and safe handling, not standalone SHM structural theory in this summer-school implementation

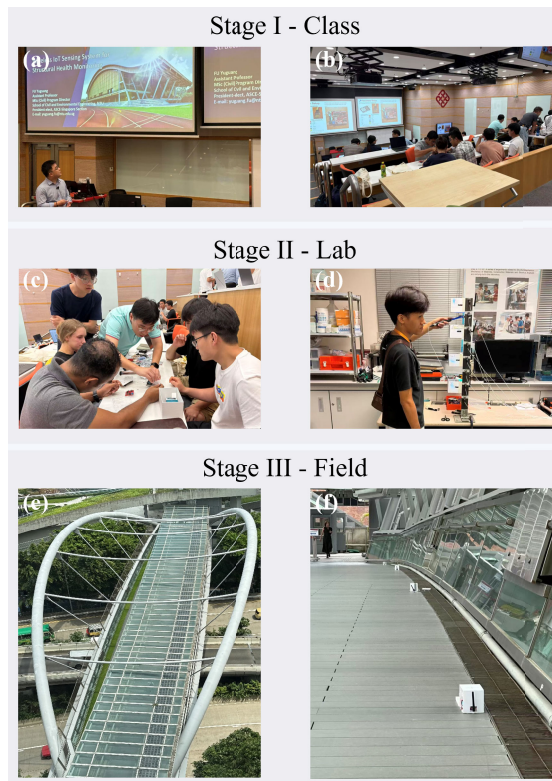


Fig. 2. CLF learning framework implementation for APESS2025 (concrete instantiation of Fig. 1). (a) Classroom instruction on sensor hardware, programming, and operations, (b) group-based learning organization, (c) sensor assembly and hardware preparation, (d) laboratory testing on a structural model with calibration procedures, (e) field deployment site at HK PolyU footbridge, and (f) actual sensor deployment and data collection.

(Section II). Inertial measurement unit (IMU) signal paths, sampling, buffering, synchronization, and logging are tied to code structure, so Lab work targets internal behavior rather than plug-and-play use. Group organization supports collaborative sense-making without replacing hands-on Lab time.

2) *Lab Stage*: Lab [Fig. 2(c) and (d)] moves to a structural test model. Teams operate the ArduinoNode WSN while synchronized high-precision acquisition records the same events for parameter calibration and for rehearsing acquisition workflows (Section II). Partially obscured acquisition code still requires students to connect principles to implementation, reconstruct missing parameters, and discuss traces before full solutions appear (Section II). Each group runs one node; eight groups plus an instructor coordinator node form the WSN used in Field. The same template is scalable to other cohort sizes: adding teams adds nodes to the shared network, so class-wide coordination practice grows with enrollment without requiring every team to duplicate a full multinode deployment kit.

3) *Field Stage*: Field [Fig. 2(e) and (f)] deploys on the PolyU butterfly-arch footbridge [34]. Students practice real deployment: sensor placement, radio or link loss, data gaps, and coordination with other systems, alongside site assessment and multisystem acquisition (ArduinoNode, phones, commercial SHM). Cross-system comparison remains an explicit analytic demand: teams interpret divergent spectra and cost-fidelity tradeoffs under ambient noise, linking acquisition choices to identification outcomes (Section II).

Following the staging in Section II, APESS2025 at Hong Kong Polytechnic University implemented CLF as a compressed module within the host summer school: ~3 h Class, ~3 h Lab, then two Field days at the footbridge, approximating a progression from classroom concept through bench work to the site even though it is not a full-semester course. Fifty-six participants in eight groups of seven collaborated on group projects using ArduinoNode alongside smartphones and commercial SHM sensors for comparison; each group assembled one ArduinoNode, and the cohort's nodes were operated jointly as one wireless sensor network in Lab and Field. Progressive skill development spans sensor selection, integration, acquisition, and deployment (capabilities often underrepresented in SHM curricula yet central to practice). Departments could expand the same sequence across multiple weeks, including additional SHM theory in Class if contact hours allow.

D. Formative Implementation, Replication, and Transition to the Platform

Across Lab and Field, activities increase in openness: calibration and workflow rehearsal on a structural model precede uncontrolled footbridge deployment. Formative work follows Section II: group presentations and instructor questioning ask students to account for how synchronization and acquisition settings shaped their records (analysis), compare predicted and observed behavior including dropouts or missing segments (reflection), and relate procedures to instrument behavior (practice). Teams process data, identify modes, and present comparisons while instructors give structured feedback on data quality, identification reasoning, presentation, and teamwork. No rubric scores or artifacts were archived for quantitative aggregation; Section IV therefore relies on questionnaire self-report, positioned as exploratory [26].

Institutional adoption depends on permissions, safety, weather, and instructor-to-team ratios on site, not only on hardware cost. The summer school is a proof of implementation rather than a turnkey recipe; later offerings can strengthen evidence with pre-post items, scored PSD checklists, or archived student artifacts. Lab-before-Field keeps failure modes in a supported setting before uncontrolled field noise; reconciling spectra with models in discussion parallels professional monitoring practice. Section III turns this staging into platform requirements: visibility, coordination, and logistics decide which features matter, rather than listing hardware for its own sake.

III. ARDUINONODE LEARNING PLATFORM: DESIGN AND IMPLEMENTATION WITHIN CLF

A. Platform Role and System Architecture

ArduinoNode carries the CLF progression from Section II through Class, Lab, and Field. Class emphasizes assembly, programming, and operation so practice targets internals, not opaque use; Lab and Field extend the same platform to coordinated structural tests and site deployment. The hardware centers on the Arduino UNO R4 Wi-Fi with integrated 802.11, paired with RF links and a synchronization stack summarized

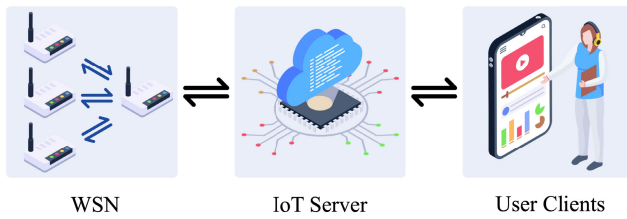


Fig. 3. Proposed ArduinoNode IoT WSN system architecture.

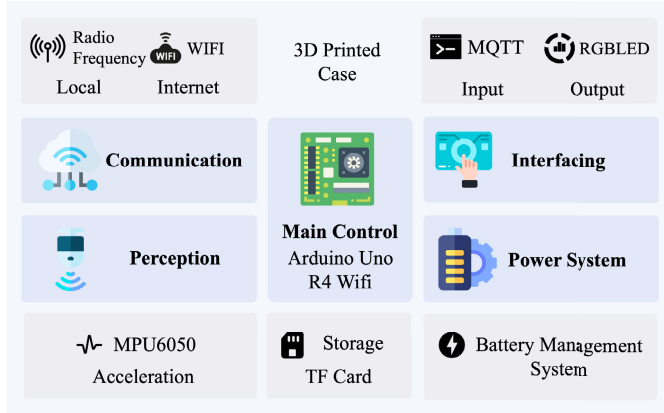


Fig. 4. Schematic illustration of ArduinoNode hardware architecture.

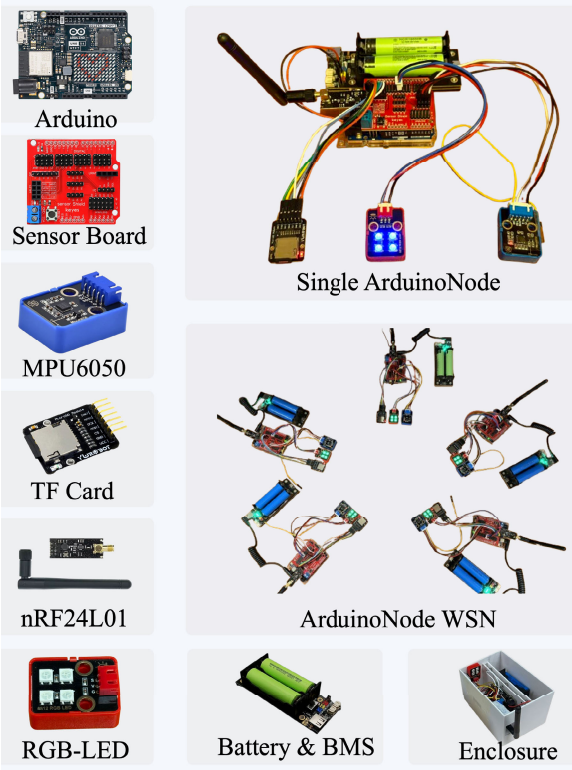


Fig. 5. ArduinoNode hardware components and network deployment.

in Section III. Design follows: manipulable sampling, storage, and communication; observable multinode coordination; student-deployable cost and robustness. Inspectable operating states and dual communication support Field exercises. Details support reproducibility; code and specifications are in the Appendix repository.

TABLE I
ARDUINONODE COMPREHENSIVE TECHNICAL SPECIFICATIONS

Parameter	Specification
Main Controller	
MCU	Renesas RA4M1
Clock Speed	48 MHz
RAM	32 KB
Flash Storage	256 KB
Acceleration Measurement	
Sensor Model	MPU6050 6-axis IMU
Measurement Range	$\pm 2/4/8/16$ g (selectable)
Resolution	16-bit
Noise Level	$0.0004 \text{ g}/\sqrt{\text{Hz}}$
Max Sampling Rate	250 Hz
Max Duration	Determined by SD storage
Wireless Communication	
WiFi Range	20m (tested)
nRF24L01 Data Rate	250 Kbps
nRF24L01 Range	25m (tested)
nRF24L01 Max Range	1100m (theoretical)
Storage	
SD Card Support	Up to 32GB
Power Supply	
Operating Voltage	5V
Battery	3.7V Li-ion (2 $\times$ 18650)

Fig. 3 shows three coupled parts: the ArduinoNode WSN, an EMQX MQTT broker [35], and desktop/mobile user interfaces for visualization and management. EMQX routes data and supports authentication, but instructional emphasis remains on edge nodes where acquisition, buffering, and local control occur. Edge-node development keeps instruction on inspectable sensing chains rather than on cloud dashboards alone.

B. Hardware Design and Specifications

Fig. 4 shows the compact integration of sensing, communication, and power for educational deployment.

1) *Subsystem and Components*: ArduinoNode combines a microcontroller, motion sensing, two wireless links, local logging, and battery power in one enclosure. The main board is an Arduino UNO R4 Wi-Fi (Renesas RA4M1) [36], i.e., a single-board MCU with integrated IEEE 802.11 for firmware and MQTT-style edge connectivity without add-on Wi-Fi shields or stackable expansion modules. Structural response uses an MPU6050 six-axis IMU with selectable $\pm 2/4/8/16$ g acceleration ranges; a Nordic nRF24L01 2.4-GHz transceiver adds peer-to-peer RF links beside Wi-Fi for mesh-style laboratory exercises. A microSD card (up to 32 GB), dual 18 650 Li-ion cells, an RGB status LED, and a 3-D-printed housing complete the assembly (Fig. 5).

2) *Key Technical Parameters*: Table I collects headline numbers for reproducibility and for Lab tuning: main-controller memory and clock; MPU6050 acceleration range, resolution, noise density, and maximum sampling rate; tested Wi-Fi and nRF24L01 ranges and data rates; SD capacity; and battery configuration. For orientation, the deployed teaching stack operates at up to 250 Hz with 16-bit samples, dual

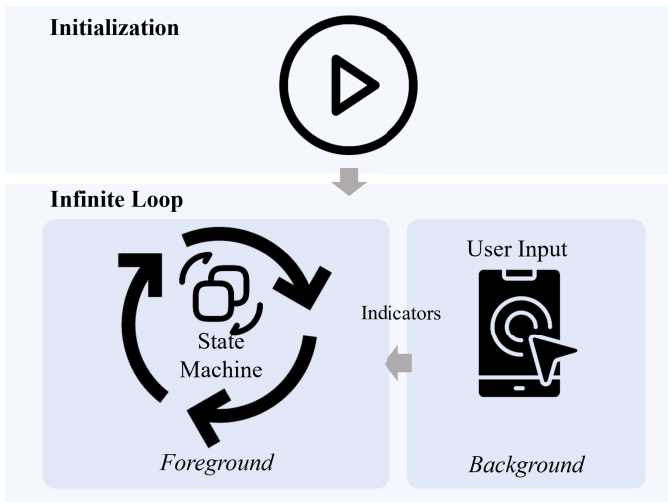


Fig. 6. Core system mechanisms: state machine and event processing.

Wi-Fi and RF links, SD-backed logging, and a typical per-node parts cost on the order of ~U.S. \$50.

3) *Educational Design Philosophy*: Open, modular hardware supports Lab scaffolding (Section II); dual communication exposes IoT and WSN idioms used in Field comparison.

C. Software Architecture, Synchronization, and Runtime Behavior

1) *Modular Architecture and Core Organization*: The firmware is organized into modules grouped as system management (configuration, state machine, time synchronization), functional processing (sensing, communication, storage), and user interface (status, commands). The layering mirrors professional embedded practice while remaining legible to learners.

2) *Core System Mechanisms*: The runtime combines three mechanisms. Initialization and steady execution follow the standard Arduino `setup()/loop()` scaffold [18], so students meet conventional embedded entry points before inspecting deeper behavior. Supervisory control uses a seven-state machine (Fig. 6) with modes BOOT, IDLE, PREPARING, SAMPLING, WI-FI_COMMUNICATING, RF_COMMUNICATING, and ERROR, binding sampling and each wireless path to explicit states instead of ad hoc branching. Event handling uses a lightweight foreground-background split: background listeners admit user or remote commands and set flags; the foreground loop consumes those flags to advance the machine, approximating predictable scheduling without adopting a full real-time operating system (RTOS).

3) *Communication and Time Synchronization*: MQTT supports remote monitoring and dashboard visibility; RF supports internode links for mesh-style exercises. Time synchronization uses NTP against internet servers at gateways (second-level) and FTSP flooding among nodes locally (millisecond-level) [21], exposing both IoT and classical WSN synchronization patterns. The combination of native Wi-Fi on the R4 module (no shield hardware), dual Wi-Fi/RF paths, and the synchronization stack yields sub-10-ms-class alignment in coordinated field capture (Section IV). NTP and FTSP follow

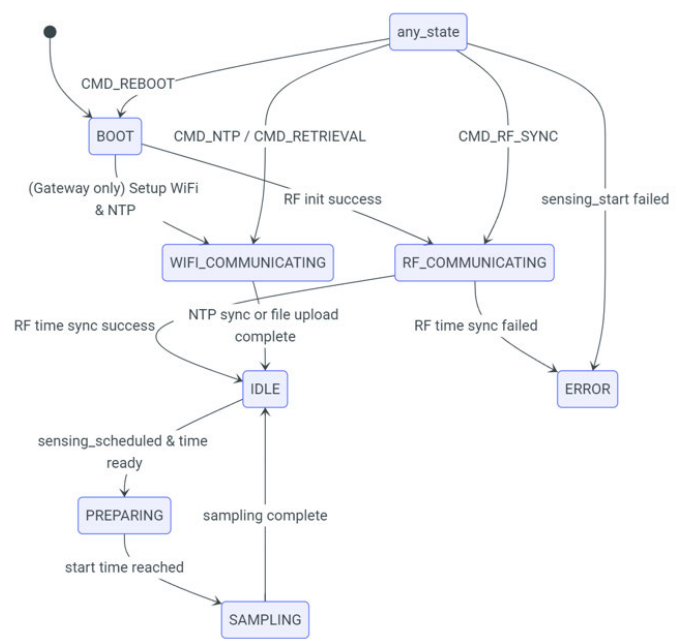


Fig. 7. State machine transition diagram.

established WSN practice; Section I argues that educational SHM literature still reports few student-built deployments pairing onboard 802.11, inspectable firmware, and comparable multinode timing at similar cost and transparency.

4) *Software Affordances and Reproducibility*: Firmware maps the limits summarized in Table I to configurable parameters that students adjust during Lab (sampling intervals, buffering, SD logging duration, and Wi-Fi versus RF paths). The Appendix repository complements that inspectability with open firmware, serial diagnostics, and instructor-facing documentation, reducing adoption friction for departments without dedicated electronics support while keeping instruction on the edge acquisition path that CLF emphasizes.

5) *State Machine Transparency and Field Coordination*: State machines structure microcontroller software through explicit modes and transitions without a full RTOS. The ArduinoNode firmware foregrounds a seven-state machine (Fig. 7) so sampling, wireless paths, and error handling map to named states rather than opaque branching, exposing control-flow behavior. Together with centralized configuration, this design supports coordinated Field exercises and keeps low-level runtime logic inspectable beyond parameter tuning.

6) *Robustness in Instructional Use*: Deployments use a star topology with a gateway center node: a simple, scalable pattern for class-scale WSN exercises. Spare nodes and light placement rules mitigate single-node RF loss; routines retry connections under marginal RF, yet links still drop as in operational WSNs, and those events are treated as instructional. If a leaf is unavailable, analyses proceed on remaining channels by omitting missing records. Discussing dropouts highlights wireless margins and hardware envelopes. Expectations target teaching feasibility, not asset-critical availability guarantees.

The frequency-domain comparison and time-domain traces are evaluated against PCB reference accelerometers. Modes are clear in the dominant (X) direction; weaker lateral compo-



Fig. 8. Laboratory and field validation. (a) Laboratory structural model with deployed sensors, (b) low-stiffness footbridge on NTU campus, and (c) high-stiffness footbridge near Nanyang Technological University campus.

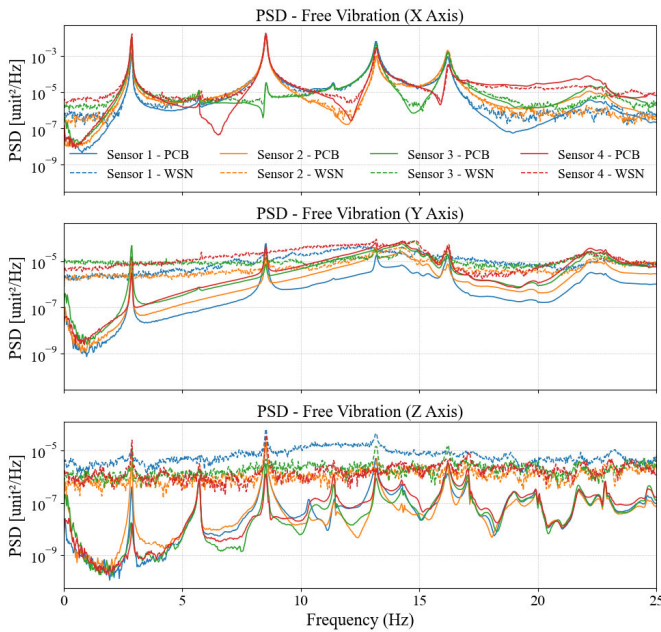


Fig. 9. Frequency-domain comparison: PSD of ArduinoNode versus PCB sensors in X-, Y-, and Z-directions.

nents still support educational system identification exercises even though they are inferior to PCB performance. Time-domain error stays within ~ 10 ms versus PCB for the compared events, adequate for the modal and timing-learning goals of Lab (see Appendix documentation for raw records and scripts).

D. Lab and Field Validation Against Reference Sensing

Validation combined laboratory structural-model tests with deployments on a low-stiffness campus footbridge and a high-stiffness highway-adjacent footbridge (Figs. 8 and 9). The model used hammer impacts and PCB reference accelerometers for comparison. The low-stiffness bridge used staged lateral manual excitation; the stiff bridge relied primarily on traffic-induced response. Together, these cases bracket stiffness

regimes students may encounter when generalizing beyond a single site.

IV. EXPERIMENTAL EVALUATION AND RESULTS

A. Study Design and Evidence Types

APESS2025 prioritized delivering CLF end-to-end with student teams. Learning experience was documented through a bilingual end-of-course questionnaire (Likert-style items on perceived improvement, engagement, and satisfaction; Appendix link). The offering was an exploratory summer-school module embedded in a host program; the compressed schedule and guest cohort limit practical scope for quantitative learning assessment beyond the postcourse survey (contact time, cohort continuity, and ethics procedures were not expanded for this deployment). Instructional emphasis concentrated on the wireless sensing chain, node firmware, timing, and coordinated field acquisition needed for CLF to run at cohort scale, rather than on parallel infrastructure for experimental learning evaluation (independent control sections, paired pre-post knowledge tests, blinded artifact scoring, or rubric aggregates archived for quantitative synthesis). Learning-outcome statements are framed as exploratory descriptive self-report, subject to social desirability and recall limits [26]. Technical measurements address whether the Field tasks assumed by CLF (modal content, multisensor contrast, coordinated acquisition) are feasible under the reported conditions. Evaluation pairs technical performance as task-feasibility evidence with questionnaire results as cohort-wide descriptive summaries of perceived experience.

Field environmental conditions on the public footbridge (wind, ambient temperature, and incidental pedestrian traffic) were not controlled; teams noted qualitative context in presentation slides rather than treating the site as a scripted modal-test laboratory. That uncontrollability is intentional: it mirrors service monitoring where excitation and noise cannot be fully prescribed, and it gives instructors concrete hooks for discussing repeatability limits when students compare PSDs across sensor classes.

B. Technical Performance Assessment

Frequency-domain summaries use Welch's method with Hann windows, 50% overlap, and length chosen per record to balance variance and resolution given each system's sampling rate and duration. Identified peaks were picked conservatively in instructor-led reviews so teams could debate leakage, harmonic contamination, and sensor-axis alignment rather than treating peak picking as fully automatic, which is consistent with the interpretive goals of Field.

Before coordinated capture windows, teams checked basic time synchronization and link health using on-node status cues so that "no data" failures became diagnosable classroom moments rather than silent gaps. SD exports were spot-checked (file sizes and timestamps) before shared post-processing to reduce copy-paste and path errors when moving from Field laptops to the analysis scripts in the Appendix.

Footbridge tests at HK PolyU [34] deployed three systems in parallel: ArduinoNode WSN (six nodes, 100 Hz, 120 s),

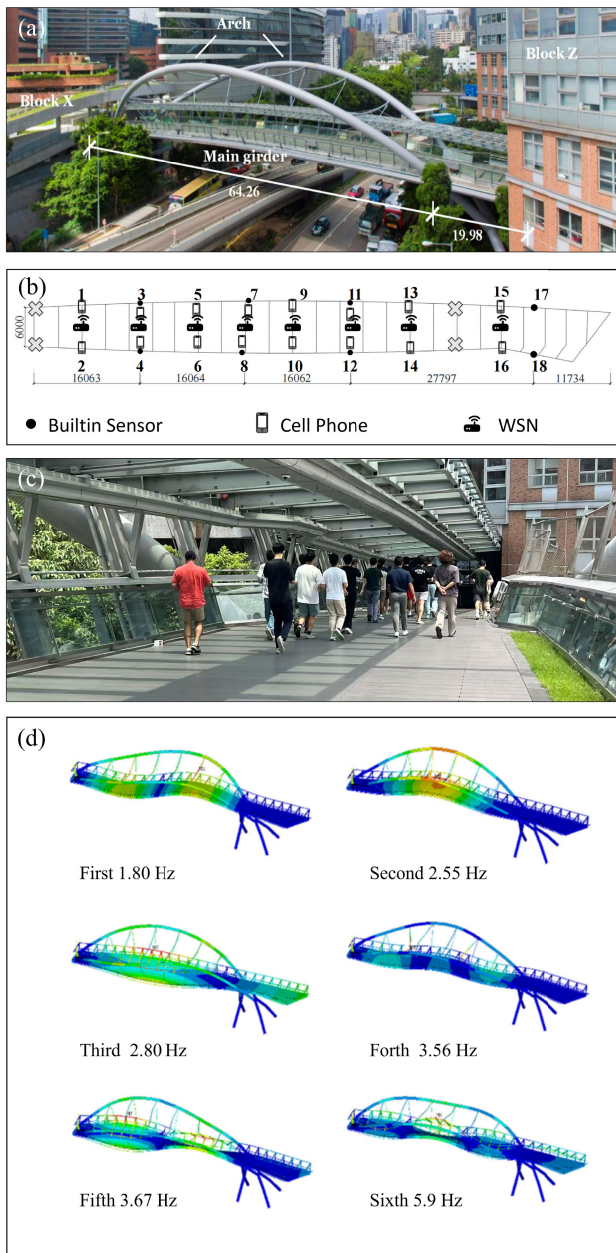


Fig. 10. HK PolyU footbridge experimental setup. (a) Bridge dimensions and structural details, (b) sensor deployment layout showing ArduinoNode WSN, mobile phone sensors, and commercial SHM system positions, (c) collective marching excitation to induce structural resonance, and (d) finite element model showing first six mode shapes and natural frequencies.

mobile-phone sensors (15 channels, 50 Hz, 430 s), and a commercial SHM array (eight accelerometers, 50 Hz, 3600 s). All records were analyzed with Welch PSDs for frequency identification. ArduinoNode achieves the sampling, resolution, and noise levels summarized in Table I; dual Wi-Fi/RF links and sub-10-ms-class synchronization support coordinated student-led acquisition. Rapid, low-cost (~U.S. \$50/node) deployment is feasible for novices, though noise exceeds research-grade sensors and occasional RF instability appears; instructors frame these tradeoffs explicitly with students as part of the learning narrative.

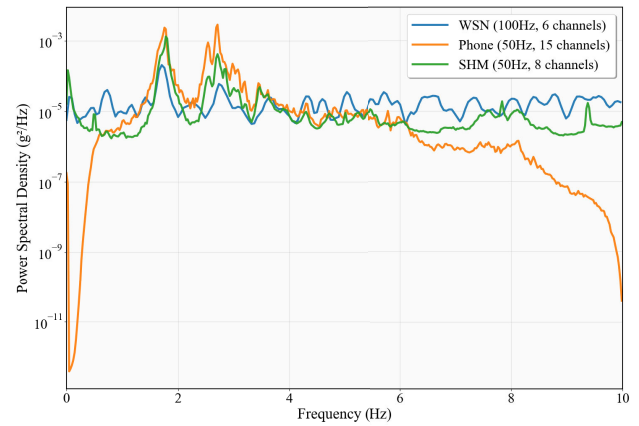


Fig. 11. PSD comparison: WSN versus Phone versus SHM.

TABLE II
IDENTIFIED NATURAL FREQUENCIES COMPARISON ACROSS
DIFFERENT SENSOR SYSTEMS

Method	Identified Frequencies (Hz)
FE Model	1.80, 2.55, 2.80, 3.56, 3.67, 5.90
ArduinoNode WSN	0.732, 1.709, 2.734, 5.029, 5.713, 8.789
Mobile Phone	1.758, 2.710, 2.979
SHM System	0.024, 1.782, 2.710, 3.076, 3.662, 7.593

Fig. 10 summarizes bridge geometry, sensor layout, collective marching excitation chosen to cooperate on low-frequency response, and a six-mode FE excerpt (1.80, 2.55, 2.80, 3.56, 3.67, and 5.90 Hz) used as interpretive targets [34]. Spatially distributed ArduinoNode, phone, and research-grade channels therefore record the same environmental variability and identification ambiguities professionals confront; that mismatch is a deliberate feature of the Field stage rather than a nuisance to erase in postprocessing.

In Table II, FE lists six modes (1.80–5.90 Hz). ArduinoNode peaks span 0.732–8.789 Hz; the sub-1-Hz component likely reflects noise. The 1.7–1.8 Hz band aligns across WSN (1.709 Hz), phones (1.758 Hz), and SHM (1.782 Hz; 5.1% error versus 1.80 Hz FE). A 2.55–2.98 Hz cluster appears on all systems (7.2% error for WSN versus 2.55 Hz). Higher WSN peaks may mix harmonics, artifacts, or local effects omitted from the six-mode FE excerpt. SHM's 0.024-Hz component illustrates sensor-class effects (a CLF teaching point). Relative quality SHM > Phone > WSN matches cost-performance expectations. Fig. 11 overlays Welch PSDs for the three footbridge systems, complementing the peak picks summarized in Table II.

C. Educational Effectiveness Evaluation

1) *Statistical Scope*: Results are descriptive percentages for the full responding cohort ($N = 56$), summarizing postmodule satisfaction and self-reported gains on the surveyed dimensions in this APES2025 offering. Broader validity limits and complementary assessment directions are discussed in Section IV-D.

2) *Survey Instrument and Cohort*: A bilingual postcourse questionnaire (Appendix) was administered after APES2025. Participation was voluntary, informed consent was obtained

before completion, and responses were collected anonymously. The instrument is organized into five thematic sections, each comprising multiple specific items; sections address demographics, perceived learning (including sensor principles, WSN architecture, and SHM/IoT-related understanding), pedagogical effectiveness, engagement with the hands-on format, and overall satisfaction (full item text in the Appendix link). Responding cohort composition: 86.7% Asia, 6.7% North America, 6.7% South America; 60% doctoral, 40% master's; and 93.3% civil engineering.

3) *Descriptive Results*: Participants reported favorable perceptions: 60% indicated substantial or strong perceived improvement in sensor working principles and SHM/IoT understanding, and 40% reported enhanced perceived understanding of WSN architecture. Engagement items were favorable (60% rating the hands-on approach as “much higher” or “significantly higher” relative to typical theory-heavy courses). Overall satisfaction was 93.3%, with 100% indicating willingness to recommend the experience. These statistics are descriptive of the APESS2025 cohort and should not be read as causal proof of learning gains.

4) *Instructor Debriefs and Future Assessment*: Informal instructor and teaching-assistant debriefs after the module consistently echoed student appreciation for seeing end-to-end data paths, troubleshooting RF dropouts collaboratively, and debating phone versus ArduinoNode spectra on the bridge, themes aligned with the CLF staging, though those field notes are not formalized or quantified here. Future implementations could archive structured reflection prompts alongside objective instruments.

D. Discussion

1) *Technical Interpretation*: The footbridge comparison supports CLF objectives: learners relate FE expectations, distributed sensing, and spectral leakage/noise tradeoffs using side-by-side systems. The 5.1% and 7.2% discrepancies on the first two FE-aligned bands are adequate for modal exercises at this price point, while spurious or extra peaks motivate instructor-led discussion of identification pitfalls, consistent with authentic complexity [29].

2) *Mechanisms (Hypothetical)*: Lab emphasizes hands-on adjustment of sampling, buffering, and synchronization; Field emphasizes interpretation under ambient noise and imperfect excitation. Multisensor PSDs (Fig. 11 and Table II) make cost-fidelity tradeoffs tangible and show how instrument class and settings shape identified spectra [27]. The questionnaire does not test these mechanisms directly; novelty and cohort selection offer plausible alternative explanations.

3) *Activity–Competency–Outcome Linkage*: Class (assembly, flashing, operation) builds sensing-chain competence; Lab (calibration, tuning, guided reconstruction) builds diagnostic/analytical competence; Field (deployment and PSD interpretation under noise/dropouts) builds transfer and judgment. Outcomes are interpreted in two matched layers: technical task feasibility and descriptive learner-reception patterns, not causal efficacy.

4) *Design Knowledge and Limitations*: The transferable unit combines sequencing logic, explicit construct-to-activity



Fig. 12. Field deployment group photograph at the Hong Kong Polytechnic University footbridge [34].

mapping (Section II), and open materials for adaptation. Internal validity for learning claims is limited (no randomized comparison, blinded scoring, or retained exam artifacts); external validity is bounded by one short program, one footbridge, and a graduate-heavy civil cohort. Technical results establish task feasibility rather than universal efficacy. Future offerings should archive rubrics or short concept checks and pursue cross-institutional replication [26], [37]. Fig. 12 documents the Field context [34].

V. CONCLUSION

SHM is an end-to-end pipeline, but instruction often emphasizes downstream modeling over how data are acquired, timed, and logged. This leaves sensing and field interpretation under-rehearsed, while commercial systems are often too closed or costly for cohort-scale inspectable practice. To address this gap, we proposed a CLF instructional architecture and ArduinoNode platform that compresses sensing-to-deployment experience from the bench to a footbridge at ~U.S. \$50 per node.

CLF progresses from Class (node assembly/programming/operation), through Lab (structural model, reference acquisition, calibration), to Field (bridge deployment, multisystem comparison, and interpretation under instructor scaffolding). It is positioned as a transferable staged design that integrates practice, reflection, and analysis, and differs from project-only, remote-lab, and maker-only formats by requiring coordinated outdoor monitoring. ArduinoNode serves as the shared artifact across stages, with inspectable firmware, dual Wi-Fi/RF links, and sub-10-ms-class synchronization for multinode capture.

Evidence combines technical feasibility with exploratory cohort reception ($N = 56$). FE-aligned bands were within 5.1% and 7.2% for the first two modes, while extra peaks supported discussion of identification artifacts. APESS2025 also reported 93.3% satisfaction, 100% willingness to recommend, and self-reported gains aligned with course emphasis (60% in sensor/SHM/IoT understanding, 40% in WSN understanding, and 60% reporting stronger hands-on emphasis than typical theory-heavy courses).

Stronger claims will require pre-post instruments, rubric-scored artifacts, performance tasks, objective measures, and replication across sites and cohorts within ethics and logistics constraints. Open code, documentation, and the questionnaire

URL are provided to support adaptation, while scaling still depends on local bridge access, safety, and assessment design.

APPENDIX

ArduinoNode GitHub Repository:

<https://github.com/Shuaiwen-Cui/ArduinoNode>

ArduinoNode Online Documentation:

<https://shuaiwen-cui.github.io/ArduinoNode/>

APESS2025 summer school Documentation:

https://shuaiwen-cui.github.io/APESS2025_ArduinoNode/

Questionnaire Link:

<https://forms.office.com/r/QZMj7Qp5zF>

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